Differentiation – Product Rule: Worked Examples

Product Rule

If $y = uv$ where *u* and *v* are functions of *x*, then

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Example

Differentiate:

$$y = (2x^2 + 4x^3)(3x^3 - 4x^2)$$

$$u = (2x^2 + 4x^3) \qquad v = (3x^3 - 4x^2)$$

$$\frac{du}{dx} = 4x + 12x^2 \qquad \frac{dv}{dx} = 9x^2 - 8x$$

$$\frac{dy}{dx} = (2x^2 + 4x^3)(9x^2 - 8x) + (3x^3 - 4x^2)(4x + 12x^2)$$

$$\frac{dy}{dx} = (18x^4 - 16x^3 + 36x^5 - 32x^4) + (12x^4 + 36x^5 - 16x^3 - 48x^4)$$

$$\frac{dy}{dx} = (-50x^4 - 32x^3 + 72x^5)$$

The result is often written in descending order

$$\frac{dy}{dx} = 72x^5 - 50x^4 - 32x^3$$

This could be factorised to

$$\frac{dy}{dx} = 2x^3(36x^2 - 25x^3 - 16)$$